

From Curiosity to Creation: Build Your First AI Project & Research Portfolio

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DAY 4 - Vibes, Aesthetics & Creative AI

Summer 2026

Discriminative vs. Generative AI

Discriminative Models (The Past)

These models learn the boundary *between* classes. They are used for classification and prediction.

- **Input:** An image of a dog.
- **Output:** The label "Dog".
- **Goal:** $P(Y|X)$ - Predict the label given the data.

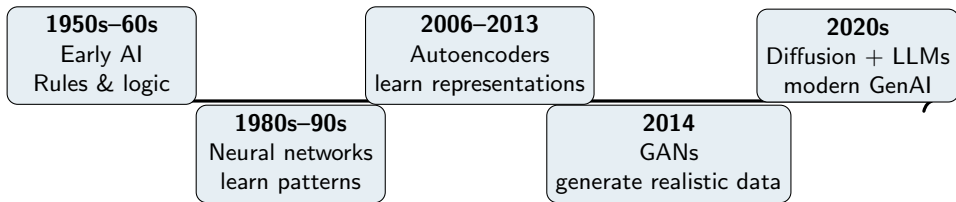
Generative Models (The Present)

These models learn the underlying distribution of the data itself. They create entirely new data.

- **Input:** The word "Dog".
- **Output:** A brand new, photorealistic image of a dog that has never existed.
- **Goal:** $P(X|Y)$ - Generate the data given the label.

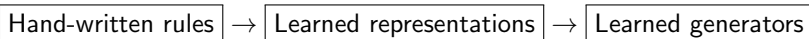
A Brief History of Generative AI

How did we get from simple models to today's generative AI?



The Big Pattern

Generative AI evolved from:



Each generation gave the machine a better way to learn what the data looks like.

From Rules to Learned Generators

1. Rule-Based AI

Human → Rules → Computer

Example:

"If the pixels look like this, call it a cat."

2. Predictive AI

Data → Learn Rules → Prediction

Example:

"Is this image a cat or a dog?"

3. Generative AI

Data → Learn Distribution → New

Example:

"Create a new image of a cat."

The Big Question

Instead of asking only

"Which category is this?"

we ask:

"What does this kind of data look like?"

and then use what was learned to create something new.

Hands-On Practice 1: Generative or Discriminative?

Mission: You are an AI detective.

For each task, decide whether the AI is mainly:

D = Discriminative

G = Generative

- ① Look at a photo and decide whether it contains a cat.
- ② Write a brand-new story about a robot living on Mars.
- ③ Look at an email and decide whether it is spam.
- ④ Create a new picture of a dragon playing basketball.
- ⑤ Given a patient's information, predict whether a disease is present.

Try It With ChatGPT

Ask ChatGPT:

"Classify these five tasks as generative or discriminative. Explain your reasoning."

Challenge: Invent one additional example of each type.

Practice 1: Answer — Generative or Discriminative?

Task	Type
Detect a cat in a photo	Discriminative
Write a Mars robot story	Generative
Detect spam	Discriminative
Create a dragon picture	Generative
Predict whether disease is present	Discriminative

The Simple Rule

Discriminative AI: “Which category does this belong to?”

Generative AI: “What new thing can I create?”

The key difference is the **output**: classification versus newly generated content.

Bonus: A chatbot that writes a poem is generative; a system that decides whether a review is positive or negative is discriminative.

Can a Machine Be Creative?

The Human View of Creativity

- We view art, music, and storytelling as deeply emotional, subjective human experiences.
- Is an AI creating art, or just committing high-tech plagiarism by blending its training data?

The Algorithmic View

"Creativity is just connecting things." - Steve Jobs.

If creativity is simply taking two existing concepts and interpolating a novel bridge between them, neural networks are mathematically perfectly suited for it.

Hands-On Practice 2: Can AI Be Creative?

Mission: Test whether an AI can combine ideas in a surprising way.

Ask ChatGPT:

"Invent 5 unusual products that combine two unrelated things.

Example: A backpack + a refrigerator = a backpack that keeps your lunch cold.

Make your ideas useful and surprising."

Now choose the **most creative** idea.

Rate it from 1–5 on:

Criterion	Score
Originality	1–5
Usefulness	1–5
Surprise	1–5

Human challenge:

Invent your own idea before looking at ChatGPT's best idea.

Practice 2: Answer — What Counts as Creative?

There is **not one correct creative idea**.

A good answer should satisfy at least three properties:

- ① **Novel:** It is not an obvious copy.
- ② **Useful or meaningful:** It does something interesting.
- ③ **Surprising:** The combination is unexpected.

For example:

Umbrella + Solar Panel

could become an umbrella that collects sunlight and charges a phone while you walk.

Important Lesson

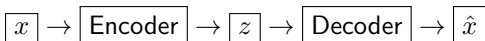
AI can generate unusual combinations of ideas. But the experiment does **not** prove that the AI has human feelings, intentions, or inspiration. It gives us a useful research question:

Is creativity the generation of novel combinations?

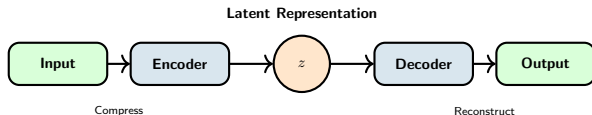
Autoencoders: Compress, Then Reconstruct

The Basic Idea

An autoencoder compresses an input into a smaller representation and then reconstructs it.



- x : original image
- z : compressed representation
- $\hat{x}$: reconstructed image



Why Is This Interesting?

The model must preserve the most important information in z .

Thus, z can become a useful mathematical representation of the original image.

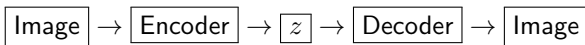
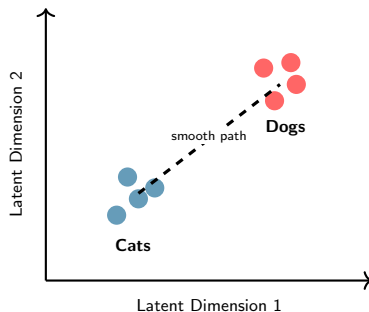
Variational Autoencoders: A Generative Latent Space

The Problem with a Basic Autoencoder

A basic autoencoder may learn a messy latent space where similar images are not arranged smoothly.

The VAE Idea

Instead of mapping each image to one exact point, a VAE learns a **small region** of possible points.



Sample $z \rightarrow$ Decode $z \rightarrow$ Generate

Big Idea

A VAE tries to organize the latent space so that **nearby points produce similar outputs**.

Smooth latent space $\Rightarrow$ easy to generate new examples

GANs: Generator vs. Discriminator

Generative Adversarial Network (GAN)

A GAN uses two neural networks with different jobs.

Generator

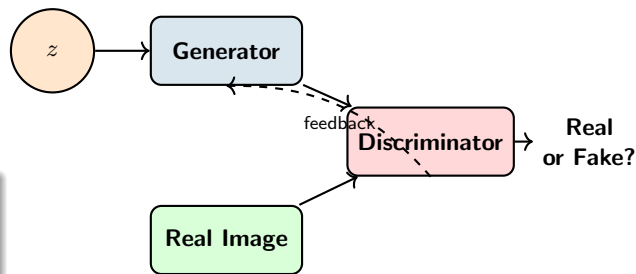
Creates fake examples.

$$z \rightarrow G(z) \rightarrow \text{Fake}$$

Discriminator

Decides whether an example is real or fake.

$$x \rightarrow D(x) \rightarrow \{\text{Real, Fake}\}$$



The Competition

Generator: **fool the detective**

Discriminator: **catch the fake**

Better Generator $\iff$ Better Discriminator

How Does a GAN Learn? Think of a Game!

Imagine an art forgery competition.

Round 1 — Beginner

Generator creates: “obviously fake”

Discriminator:

Easy to detect!

Round 2 — Improvement

Generator becomes better.

Discriminator also becomes better.

Harder to detect!

Generate → Detect → Improve → Generate Again

Goal

Eventually, Generator creates images that are difficult for Discriminator to distinguish from real images.
This adversarial competition produced remarkably realistic synthetic images.

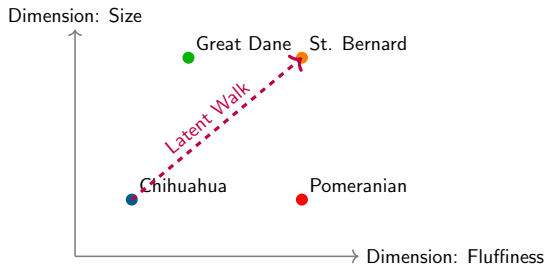
Think: What happens if the Generator gets too good? What happens if the Discriminator gets too good?

What is the Latent Space?

The Vibe Landscape

Generative models compress all human visual concepts into a massively high-dimensional geometric map called the **Latent Space**.

- It is a coordinate system of concepts.
- Every possible image is a single $(x, y, z, \dots)$ coordinate in this space.
- If you move slightly to the left, the image might get "happier." If you move up, it might get "more photorealistic."



Latent Interpolation: Math Becomes Art

Vector Algebra with Images

Because the Latent Space is continuous, we can pick any two points and mathematically walk between them.

- Point A: A photograph of an apple.
- Point B: A photograph of a dog.
- The exact midpoint $(A + B)/2$ is a smooth, AI-generated hybrid "apple-dog."

Prompting is just GPS navigation

When you type a prompt into Midjourney or DALL-E, you are not telling it how to draw. You are giving the algorithm a set of GPS coordinates to locate a specific point in the Latent Space, which it then renders into pixels.

Hands-On Practice 3: Take a Walk in Latent Space

Mission: Imagine that two ideas are two points on a map. Suppose:

$$\text{Happy Image} = A = (2, 8)$$

and

$$\text{Sad Image} = B = (8, 2).$$

The halfway point is:

$$M = \frac{A + B}{2}.$$

Your Challenge

Calculate the midpoint M . Then imagine a 5-step journey:

$$A \rightarrow \boxed{?} \rightarrow \boxed{?} \rightarrow \boxed{?} \rightarrow B$$

What might the images look like as we gradually move from “happy” to “sad”?

Ask ChatGPT: “Describe five images that gradually interpolate from a very happy to a very sad scene.”

Practice 3: Answer — Latent Interpolation

We have

$$A = (2, 8), \quad B = (8, 2).$$

Therefore:

$$M = \frac{(2, 8) + (8, 2)}{2} = \frac{(10, 10)}{2} = \boxed{(5, 5)}.$$

A simple conceptual path might be:

$$(2, 8) \rightarrow (3.5, 6.5) \rightarrow (5, 5) \rightarrow (6.5, 3.5) \rightarrow (8, 2)$$

Big Idea

If nearby locations in a latent space correspond to related concepts, moving gradually through the space can create gradual changes in the generated result.

Start $\rightarrow$ Move $\rightarrow$ New concept

This is the intuition behind **latent interpolation**.

From GANs to Diffusion: A New Strategy

GANs

Learn to generate by playing a competition:

Generator $\leftrightarrow$ Discriminator

- Powerful
- Can produce sharp images
- Training can be difficult

Diffusion Models

Learn to generate by learning how to **remove noise**:

Noise $\rightarrow$ Less Noise $\rightarrow$ Image

- Stable training
- High-quality generation
- Excellent control

The Big Evolution

Autoencoder $\rightarrow$ VAE $\rightarrow$ GAN $\rightarrow$ Diffusion

Different models use different mathematical strategies, but they all try to answer the same question:

How can a machine learn what realistic data looks like?

How do we generate the pixels? (Diffusion Models)

Modern image AI (Stable Diffusion, DALL-E 3) relies on a physics concept called **Diffusion**.

Phase 1: Forward Diffusion (Destruction)

During training, the AI takes a clean image of a cat and slowly adds static (Gaussian noise) to it over 1,000 steps until it is completely unrecognizable TV static.

Phase 2: Reverse Diffusion (Creation)

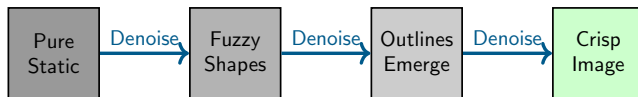
The neural network is trained to do the exact opposite. We give it pure static, and teach it to subtract the noise, step-by-step, until a clean image emerges.

Visualizing the Diffusion Process

From Pure Noise to Structured Art

Text Conditioning Guides the Noise Removal

Text Prompt: "A cute dog"



Hands-On Practice 4: Be the Diffusion Model

Mission: Imagine you are an image-generating AI. A clean image begins as: CAT. Then noise is gradually added:

CAT $\rightarrow$ CAT + noise $\rightarrow$ very noisy $\rightarrow$ almost random

Now the AI must work backwards.

Your Challenge

Imagine you see this sequence:

Random Noise $\rightarrow$ fuzzy ears $\rightarrow$ cat-shaped outline $\rightarrow$ cat

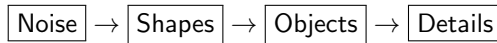
At each step, answer:

- 1 What information is being recovered?
- 2 Why is the final step easier than starting from pure noise?
- 3 What would happen if the model removed too much noise?

Optional: Ask ChatGPT to explain diffusion using a “sculptor carving a statue” analogy.

Practice 4: Answer — How Diffusion Creates Images

As the model removes noise, increasingly useful structure appears:



1. What is recovered?

Edges, shapes, textures, and eventually recognizable objects.

2. Why do later steps help?

Once rough structure appears, the model has more information about what final image should look like.

3. What if too much noise is removed incorrectly?

The final image may contain distorted or incorrect features.

Big Idea

Diffusion generation is not:

“Draw the entire picture at once.”

Instead, it is approximately:

Start with noise and repeatedly make it more structured.

Neural Style Transfer (NST)

Separating Content from Style

AI researchers discovered that Convolutional Neural Networks (CNNs) process images in a hierarchy that allows us to mathematically separate *what* is in an image from *how* it is drawn.

- **Content:** The physical objects (e.g., a house, a tree). These are recognized in the *deep* layers of the network.
- **Style:** The brushstrokes, colors, and textures. These are captured in the *shallow* layers.

The Formula for Art

By combining the Content vector of a photograph with the Style vector of a painting (e.g., Van Gogh's *Starry Night*), the network optimizes a new image that perfectly blends both.

Textual "Vibe Transfer"

We can apply the exact same concept to language using Large Language Models. We can separate the **semantic content** (the facts) from the **syntactic style** (the vibe).

- *Base Content*: "The AI model achieved 95% accuracy on the test set."
- *Pirate Style*: "Arrr, the metal brain conquered 95% of the treacherous exam!"
- *Academic Style*: "Empirical evaluation demonstrates a 95% top-1 accuracy metric across the validation distribution."

The Mechanics of Vibe

This works because the LLM traverses the latent space. By anchoring the generation to a specific contextual cluster (e.g., 18th-century nautical terms), it forces the output probabilities to align with that aesthetic.

Multi-Agent Systems: AI Talking to AI

A single AI model has limitations. It gets stuck in loops or hallucinates without realizing it.

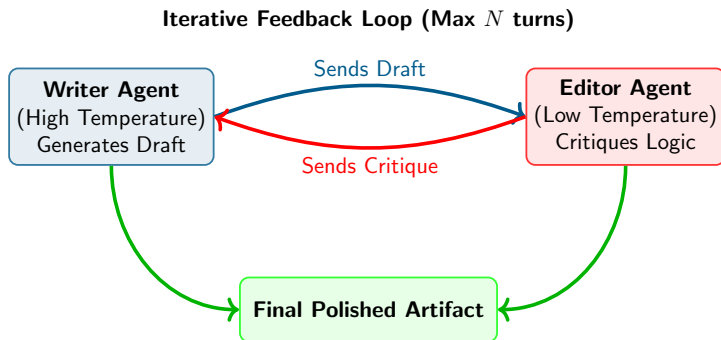
The Solution: Multi-Agent Frameworks

- Instead of using one LLM, we initialize two (or more) separate AI personas and force them to talk to each other.
- **Agent 1 (The Generator):** Tasked with brainstorming and creating raw ideas.
- **Agent 2 (The Critic):** Tasked with finding logical flaws, plot holes, or aesthetic errors.

Emergent Creativity

By looping their conversation, the system self-corrects. The final output is vastly superior to what a single agent could produce on its own.

Visualizing the Collaborative Loop



Hands-On Practice 5: Build an AI Creative Team

Mission: Can two AI “agents” make a better story than one?

Open two ChatGPT conversations.

Agent 1 — The Writer

Give it:

“Write a short story about a robot who wants to become a chef. Make it funny and surprising.”

Agent 2 — The Critic

Copy the story into the second conversation and ask:

“You are a strict editor. Find three problems in this story and give three specific suggestions for improvement.”

Your Task:

Return to the Writer and ask it to revise the story using the Critic’s feedback.

Question: Is the final story better?

Practice 5: Answer — The AI Creative Team

There is no single “correct” story. Instead, compare:

Draft 1

vs.

Revised Draft

Use three criteria:

- 1 Is the plot easier to understand?
- 2 Is the story more interesting?
- 3 Did the revision actually fix the Critic’s problems?

What Should You Discover?

The Critic can notice problems that the Writer did not notice.

The Writer can then use that feedback to generate a better draft.

Generate → Critique → Revise → Generate Again

This is the basic idea behind a **multi-agent creative loop**.

Research question: Does more feedback always make the story better?

Generative AI Safety & Ethics

With the ability to perfectly synthesize human text, audio, and images, Generative AI introduces massive ethical risks.

- **Deepfakes:** Non-consensual generation of realistic photos/audio of real people.
- **Copyright:** Generative models are trained on billions of scraped images from human artists who were not compensated.
- **Misinformation:** LLMs can generate fake news articles at infinite scale, complete with fabricated citations.

The Alignment Problem

How do we program human values into a matrix of floating-point numbers? How do we ensure the model refuses dangerous prompts (e.g., "How to build a bomb") without censoring harmless creativity?

The Irony of Creativity

"Creativity is just highly-rewarded hallucination."

When we want facts:

If you ask an AI for the capital of France and it says "A shimmering city of glass," that is a **hallucination**. It is a failure of the system.

When we want art:

If you ask an AI to write a fantasy novel and it describes "A shimmering city of glass," that is **creativity**. It is a success.

Mechanistically, the AI is doing the exact same mathematical operation in both scenarios.

Hands-On Practice 6: Creativity or Hallucination?

Mission: Decide whether each AI response is a **creative success** or a **hallucination**.

- ① Prompt: "Write a fantasy story about a city floating above the clouds."
AI: "The city was built from silver glass and powered by the dreams of its citizens."
- ② Prompt: "What is the capital of France?"
AI: "Paris, a city floating above the clouds."
- ③ Prompt: "Invent a fictional planet."
AI: "Zorax has three moons and oceans made of glowing blue liquid."

For each, write:

C = Creative

H = Hallucination

Then ask ChatGPT:

"Explain why each answer is creative or a hallucination."

Practice 6: Answer — Creativity or Hallucination?

Case	Answer	Why?
1	C	Fiction was requested
2	H	The factual answer is wrong
3	C	A fictional planet was requested

The Important Insight

The underlying language-generation mechanism can be similar in all three cases. What changes is the **task**.

Creative invention is good when invention is wanted.

The same behavior is a hallucination when facts are wanted.

Research question:

How could we design a prompt or AI system that knows when to **invent** and when to **verify**?

PhD-Led Lab 1: Vibe Transmutation (1:00 PM)

Image Style Mixing

- We will access a generative image pipeline.
- You will experiment with prompt weighting (e.g., `dog:1.5`, `cyberpunk:-0.5`) to mathematically navigate the latent space.
- Goal: Force two conflicting concepts to merge seamlessly.

Text "Vibe Transfer"

- We will take a dry, factual Wikipedia article.
- You will engineer prompts that force the LLM to rewrite the article in increasingly bizarre constraints (e.g., as a Shakespearean sonnet, as 90s hacker slang).

PhD-Led Lab 2: Multi-Agent Story Creation (2:40 PM)

Building the Collaboration Loop

- You will programmatically link two LLM instances using Python.
- **Agent A** will be instructed to write a short story.
- **Agent B** will be instructed to ruthlessly critique Agent A's story.
- You will run the loop 3 times and analyze how the artifact evolves.

The Output

This lab forms the foundation for advanced AI architectures. You are no longer just prompting; you are building automated software ecosystems.

Your Goal for Today (4:00 PM Check-In)

The Deliverable

You must finalize your **Core Project Prototype**.

- *Visual*: Generate the final artifacts or images for your portfolio.
- *Narrative*: Output the final text from your multi-agent system.
- *Interactive*: Ensure your code/chatbot runs without crashing.

Tomorrow is the Showcase

Tomorrow, you will compile everything (your research question, your interpretability explanation, your philosophical reflection, and today's final artifact) into a formal presentation.

Thank you!

Questions?

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